Approaching Wealth Inequality with Subdiffusion Processes

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Abstract

Standard Bewley-Huggett-Aiyagari (BHA) heterogeneous-agent models assume that individual labor income follows a Markov process with Gaussian innovations, implying that agents are memoryless and that income variance grows linearly in time. However, a growing empirical literature documents that earnings dynamics are fundamentally non-Gaussian and non-Markovian, exhibiting heavy tails, asymmetric persistence, and long memory. This proposal introduces subdiffusion into the BHA framework by recasting the income process as a Continuous Time Random Walk (CTRW), in which agents pause for heavy-tailed waiting times between transitions. At the macroscopic level, this yields a variational time-fractional Mean Field Game (MFG) system consisting of a backward time-fractional Hamilton-Jacobi-Bellman equation and a forward time-fractional Fokker-Planck equation, following the framework of Qing and Camilli (2018). I propose to solve this system using the continuous-time finite difference approach of Achdou et al. (2022), adapted for fractional operators, and implemented via the MFAX framework of Wibault et al. (2026). The model is calibrated against the Survey of Consumer Finances and Social Security Administration earnings data, with the subdiffusion exponent identified from the kurtosis-horizon relationship documented in Guvenen et al. (2015).

1 Introduction

Singular boundary value problems (BVPs) are often difficult to solve because the nature of the solution near the singular point is unknown—it could be regular, a pole, or a logarithmic singularity. In economics, when solving for value functions in heterogeneous-agent models, various methods have developed in something of a convergent evolution, with cubic splines and general optimal control methods emerging in parallel to the physics and applied mathematics literature. Drawing upon this wealth of cross-disciplinary methodology (Tarasov 2020), this proposal argues that the income process itself—not just the numerical method—can be enriched by importing ideas from the physics of anomalous diffusion.

The workhorse framework for studying wealth inequality is the Bewley-Huggett-Aiyagari (BHA) model (Huggett 1993; Aiyagari 1994), in which a continuum of infinitely-lived households accumulate precautionary savings against uninsurable idiosyncratic labor income risk. When recast in continuous time, the equilibrium of this model is characterized by a coupled system of a Hamilton-Jacobi-Bellman (HJB) equation for the individual value function and a Kolmogorov Forward (KF) equation for the cross-sectional distribution of income and wealth (Achdou et al. 2022). Despite substantial advances in the quantitative performance of these models, a fundamental limitation remains in the specification of the underlying income process. The standard formulation treats individual labor income as a Markov process with Gaussian innovations, typically an AR(1) discretized via Tauchen's method (Tauchen 1986). This assumption implies that agents are *memoryless*—the future distribution of income depends only on the current income state, not on the path that led there—and that income uncertainty grows linearly in time, as in standard Brownian motion. Neither implication is consistent with the microeconomic data (Guvenen et al. 2015).

I propose to address this limitation by introducing a subdiffusive income process into the BHA framework. Intuitively, instead of income evolving continuously, agents "pause" for random waiting times between income states, where these waiting times are governed by a heavy-tailed power law. This means that some agents are trapped in low-income states for far longer than a standard Poisson process would predict—capturing unemployment spells with fat-tailed durations, careers with long periods of stagnation punctuated by sudden jumps, and job-market lock-in. The resulting model is a variational time-fractional Mean Field Game, following Qing and Camilli (2018) and the earlier founda-

tional work of Camilli and Maio (2019). To my knowledge, no prior work has embedded a subdiffusive income process into a BHA general equilibrium model in this way. The closest parallel is Gabaix (2016), which highlights that scale-dependent earnings drive top-tier income inequality, but does not consider fractional dynamics or non-Markovian waiting times. Previous work with fractional calculus and subdiffusion has been touched upon in the finance literature (Benzaquen and Bouchaud 2018), but no work has been done in the macroeconomic context proposed here.

2 Review of Literature

A Modeling Framework

Mean Field Game (MFG) models have provided a rigorous framework for the analysis of dynamic, stochastic games involving a large number of rational and indistinguishable agents (Qing and Camilli 2018). The general theory is based on the foundational works of Lasry and Lions (2006) and Lasry and Lions (2007). The benchmark MFG model in macroeconomics is the BHA framework, which, when recast in continuous time, reduces to a coupled system of partial differential equations known as the backward-forward MFG system (Achdou et al. 2022). The HJB equation characterizes the individual agent's value function:

$$\rho\, v(a,z) = \max_{c} \left\{ u(c) + \partial_a v(a,z) \cdot (ra + wz - c) + \mathcal{L}_z v(a,z) \right\}, \tag{1}$$

where a is wealth, z is income, r is the interest rate, w is the wage, ρ is the discount rate, and $\mathcal{L}_z$ is the infinitesimal generator of the income process. The KF equation describes the evolution of the joint density $g(a,z,t)$:

$$\partial_t g = -\partial_a \left[\mu^*(a,z)\, g \right] + \mathcal{L}_z^* g, \tag{2}$$

where $\mu^*(a,z) = ra + wz - c^*(a,z)$ is the optimal drift and $\mathcal{L}_z^*$ is the adjoint of $\mathcal{L}_z$. A key computational advantage of this continuous-time framing is that the discretized KF matrix is the transpose of the HJB transition matrix, allowing the population distribution to be computed essentially for free once the individual problem is resolved (Achdou et al. 2022). Furthermore, this approach naturally generates sparse matrices, which are essential for handling large-scale models where approximate aggregation may not hold.

A notable extension is provided by Krusell and Smith (1998), which incorporates common noise in the form of aggregate productivity shocks. Under full rational expectations, this results in a "Master Equation" where the entire infinite-dimensional cross-sectional distribution becomes a state variable, as agents must forecast its evolution to predict future prices (Moll 2025).

An important extension of the standard MFG is the variational formulation, based on optimal transport (Benamou and Brenier 2000). Cardaliaguet et al. (2015) obtained existence and uniqueness of the MFG system using local coupling via variational methods. The distinction between a regular MFG and a variational MFG lies in whether the equilibrium is characterized by a system of equations or as the optimality condition for a global optimization problem. This is particularly useful in time-fractional systems, where the variational structure helps characterize non-Markovian dynamics and long-memory effects that are difficult to handle with standard methods (Camilli and Maio 2019). Other models used to study inequality have also used this framework, for example by adding Pareto tails (Hubmer, Krusell, and Smith. 2021).

In particular, I wish to consider a variational time-fractional mean field game to more closely represent real income earnings dynamics by incorporating a subdiffusion process (Qing and Camilli 2018).

B Current Labor Income Modeling

In quantitative models, labor income is formulated as a Markov chain or general diffusion process of the form

$$dz_t = \mu\, dt + \sigma\, dW_t, \tag{3}$$

where W_t is a standard Brownian motion, and then discretized according to Tauchen's method (Tauchen 1986; Tauchen and Hussey 1991) or Rouwenhorst's method (Rouwenhorst 1998). This model has linear growth and normal diffusion: $\mathbb{E}[(z_t - z_0)^2] = \sigma^2 t$.

More recent research has incorporated Pareto tails and superstar states to better match the empirics of the very wealthy (Castañeda, Díaz-Giménez, and Ríos-Rull 2003). Other approaches involve discretizing the entire earnings distribution into percentiles, effectively creating a step function where each step is the average earnings of that bin (De Nardi and Fella 2017). This incorporates some persistence of the earning process but maintains the assumption that la-

bor income is memoryless and does not admit heterogeneity in waiting times. Methods such as Hidden Markov Models (Zhou 2023) allow for non-Gaussian transitory and persistent components, but still maintain the Markovian assumption: future income depends only on current income and possibly a hidden state, not on the history of waiting times. However, recent literature has posited that labor income does not strictly follow (3), as models following this specification do not precisely match the data (Guvenen et al. 2015). Different careers have different earnings structures, and agents are more likely to experience discrete jumps in pay than continuous changes.

I propose that introducing a subdiffusion process into the BHA model will not only better represent these empirical facts, but will also enable us to draw on a rich body of pre-existing mathematical results for analysis.

C Subdiffusion

A subdiffusion process is a stochastic framework that characterizes systems where the movement of variables is slower than standard Brownian motion, fundamentally departing from standard Markovian models where agents are memoryless and future states depend only on the current position (Camilli and Maio 2019). A standard Brownian income process is as in (3), with linear growth and normal diffusion $\mathbb{E}[(z_t - z_0)^2] = \sigma^2 t$. Subdiffusion replaces this with

$$\mathbb{E}\left[(z_t - z_0)^2\right] \sim K_\beta \, t^\beta, \qquad 0 < \beta < 1, \tag{4}$$

where K_β is a generalized diffusion coefficient (**metzlerRandomWalkGuide2000**). This is slower than normal diffusion because the process is trapped for long durations at each state.

On a microscopic level, these dynamics are governed by a CTRW in which an individual agent pauses for random waiting times distributed according to a heavy-tailed power-law (Qing and Camilli 2018):

$$\psi(\tau) \sim C \, \tau^{-(1+\beta)}, \quad \tau \to \infty.$$

This mechanism introduces a distinction between external clock time and an internal time scale modeled as an inverse stable subordinator, wherein the agent's internal clock effectively stops during a trapping event. The resulting income process is $z_t = z_{Y_{\mathcal{E}_t}}$, a time-changed Markov chain where Y_s is the underlying chain and $\mathcal{E}_t = \inf\{s > 0 : L_s > t\}$ is the inverse of a β-stable subordinator

L_s. Because the probability of an agent's next move depends on the duration they have already spent in their current state, the process lacks stationary and independent increments in the individual trajectory, effectively curing the amnesia of Markovian models by making future dynamics dependent on the system's entire history (Tarasov 2020). It is important to note, however, that this non-stationarity of individual increments does not preclude the existence of a stationary cross-sectional distribution $g^*(a, z)$ at the population level; under the Riemann-Liouville formulation adopted below, Camilli and Maio (2019) establish that the MFG system admits a well-posed solution, and the long-run population distribution differs from the standard Markovian case through the fractional weighting of the operator $\mathcal{L}_z^*$.

Macroscopically, these subdiffusive regimes are described by a time-fractional Fokker-Planck equation, in which the standard time derivative ∂_t is replaced by a Reimann-Louisville fractional derivative ∂_t^β of order $\beta \in (0, 1)$:

$$\partial_t^\beta g = -\partial_a[\mu g] + \frac{\sigma^2}{2}\partial_{aa} g.$$

The Reimann-Loisville fractional derivative is chosen (as opposed to the Caputo fractional derivative) following the literature and it being slightly better for studying the distributional component. However, Qing and Camilli 2018 prove the same result with Caputo fractional derivatives so its certainly an avenue worth exploring. Fractional derivatives derivative $\partial_t^\beta g(t) = \frac{1}{\Gamma(1-\beta)} \int_0^t (t-s)^{-\beta} g'(s)\, ds$ is a nonlocal operator: the current rate of change of the density g depends on its entire history, encoding the memory of the CTRW waiting times (Tarasov 2020). In the economics and finance literature, this non-Markovian structure is instrumental for modeling long-memory effects and trading latency (Benzaquen and Bouchaud 2018).

3 Methodology

A The Model

We start with the BHA model and recast it into the MFG framework consisting of an HJB and a KF equation. Consider a continuum of infinitely-lived households, each characterized by wealth $a \in [\underline{a}, \infty)$ and labor income z. The

household's wealth evolves according to:

$$\dot{a}_t = ra_t + wz_t - c_t, \tag{5}$$

where r is the interest rate, w is the wage, and $c_t \geq 0$ is consumption, subject to the borrowing constraint $a_t \geq \underline{a}$. In the standard BHA model, agents' wealth evolves via a drift-diffusion process or a discrete Markov chain. To incorporate subdiffusion, I assume that at the microscopic level, an individual's income is governed by a CTRW, mathematically represented as the time-changed process $z_t = Y_{\mathcal{E}_t}$ described in Section 2.

The standard HJB equation is replaced by a backward time-fractional HJB equation. Following Qing and Camilli (2018) and Camilli and Maio (2019), the value function $v(a, z, t)$ satisfies:

$$-\partial_t v + D^{1-\beta}_{[t,T)} \left[\mathcal{H}(a, z, \partial_a v)\right] = f(a, z, g), \tag{6}$$

where $\mathcal{H}(a, z, p) = \max_c\{u(c) - p(ra + wz - c)\}$ is the Hamiltonian, $f(a, z, g)$ is the coupling to the cross-sectional distribution, and $D^{1-\beta}_{[t,T)}$ is the backward Riemann-Liouville fractional derivative of order $1 - \beta$:

$$D^{1-\beta}_{[t,T)}\phi := -\frac{1}{\Gamma(\beta)} \frac{d}{dt} \int_t^T \frac{\phi(\tau)}{(\tau - t)^{1-\beta}}\, d\tau.$$

For the stationary problem ($T \to \infty$), this fractional derivative encodes the agent's valuation of future income states, weighted by the probability of remaining active (not trapped) at each future time.

The KF equation is similarly modified to a forward time-fractional Fokker-Planck equation, describing the evolution of the joint density $g(a, z, t)$:

$$\partial_t g - \partial_a \left[\mu^*(a, z)\, D^{1-\beta}_{(0,t]} g\right] = \mathcal{L}^*_z g, \tag{7}$$

where $\mu^*(a, z) = ra + wz - c^*(a, z)$ is the optimal saving drift, $D^{1-\beta}_{(0,t]}$ is the forward Riemann-Liouville fractional derivative, and $\mathcal{L}^*_z$ is the adjoint of the income generator. This equation describes how the population distribution spreads through the state space more slowly than standard diffusion due to the persistent traps in the CTRW. The forward fractional derivative $D^{1-\beta}_{(0,t]} g$ captures the memory of the population: the current rate of change of the distribution depends on the entire history of g up to t, with weights that decay as a

power law rather than exponentially. Importantly, the fractional KF equation preserves the positivity of the density and ensures that the total mass integrates to one (Camilli and Maio 2019).

Following Qing and Camilli (2018), the system (6)–(7) can be derived as the first-order optimality condition for a pair of convex control problems, which guarantees existence and uniqueness of the equilibrium under standard convexity conditions, extending the classical results of Cardaliaguet et al. (2015) to the subdiffusive setting. Equilibrium closes with capital market clearing:

$$K = \int a\, g(a,z)\, da\, dz, \quad r = f'(K) - \delta, \quad w = f(K) - Kf'(K).$$

The model includes incomplete markets, precautionary savings, a borrowing constraint, subdiffusive income dynamics with memory, and general equilibrium through capital market clearing. Aggregate productivity shocks (Krusell-Smith), discrete choice, life-cycle structure, and multiple assets are abstracted away; these extensions are left for future work, and the stationary single-asset version is already a novel and tractable first contribution.

B Computational Methods

Following the literature, the coupled equations will be solved as a singular boundary value problem in continuous time using the method of Achdou et al. (2022), adapted to fractional operators. This approach discretizes the HJB and KF equations using an upwind finite difference scheme that automatically handles the singular boundary condition at $\underline{a}$ and produces sparse, diagonally dominant matrices. I adapt this to the fractional setting using the L1 scheme for the Caputo derivative on a time grid $\{t_0, \ldots, t_N\}$:

$$\partial_t^\beta g^n \approx \frac{(\Delta t)^{-\beta}}{\Gamma(2-\beta)} \sum_{k=0}^{n-1} b_{n-k}^{(\beta)} \left(g^{k+1} - g^k \right),$$

where $b_j^{(\beta)} = j^{1-\beta} - (j-1)^{1-\beta}$ are the L1 weights, which decay hyperbolically and encode the long-memory character of subdiffusion. The spatial discretization uses the same upwind scheme as Achdou et al. (2022), preserving the M-matrix property and guaranteeing non-negativity of the density. The nonlinearity in (6) is handled by Howard's policy iteration: given the current policy $c^n(a,z)$, one computes the upwind transition matrix $A(c^n)$ and solves the linear fractional

system $(\rho I - A_\beta(c^n))\mathbf{v}^{n+1} = \mathbf{u}^n$, before updating $c^{n+1} = (u')^{-1}(\partial_a v^{n+1})$.

I believe the reason this method has not been exploited in the current macroeconomics literature is largely computational. The L1 scheme introduces a cost of $O(N^2)$ in time rather than $O(N)$, since each new time step requires summing over all past steps. However, with recent improvements in computation, and in particular JAX's utilization of parallelization and just-in-time compilation, implementing this method for large macroeconomic problems is now feasible. Particularly relevant to solving mean field games in JAX is the introduction of MFAX (Wibault et al. 2026), a JAX-based framework for MFGs that supports both analytic and sample-based mean field updates, leverages known transition dynamics for variance reduction via the Recurrent Structural Policy Gradient (RSPG) algorithm, and crucially supports history-aware, non-Markovian policies—precisely what subdiffusive dynamics require. For the grid sizes needed for accurate distributional analysis ($N \sim 500$ time steps, $I \sim 500$ wealth grid points, $J \sim 10$ income states), total computation time is expected to be on the order of minutes on a modern GPU.

C Verification

In order to empirically validate the model, I propose two verification strategies. First, I check that the model collapses to the standard BHA solution as $\beta \to 1$, since in this limit the fractional operators reduce to their standard first-order counterparts and the CTRW reduces to a standard Poisson process. Second, I verify that the model matches the cross-sectional moments of the earnings change distribution documented in Guvenen et al. (2015): specifically the kurtosis-horizon relationship $\text{Kurt}(\Delta z_h) \sim h^{1-\beta}$, which directly identifies β, and the negative skewness of earnings changes at different income levels. These verification exercises serve both as internal consistency checks and as evidence that the model mechanism is correctly specified.

D Comparisons and Calibrations

The model will be compared with current empirics using the Survey of Consumer Finances (**SCF2022**) and the Social Security Administration earnings records as analyzed in Guvenen et al. (2015). Standard parameters—the discount rate ρ, risk aversion γ (for CRRA utility $u(c) = c^{1-\gamma}/(1-\gamma)$), and production function parameters $\{A, \alpha, \delta\}$—are calibrated to match aggregate U.S. data following Aiyagari (1994). The income state values and transition structure are calibrated

to match the mean, variance, and Gini coefficient of the earnings distribution from the SSA data. The subdiffusion exponent β is identified from the slope of kurtosis versus time horizon in the SSA data. Further adjustments to the parameters will be made as necessary to calibrate the model to the wealth distribution moments from the SCF: the Gini coefficient, the wealth shares of the top 1%, 5%, and 10%, and the fraction of households at the borrowing constraint.

Once calibrated, the model will be used to compare the stationary wealth distribution under subdiffusion versus standard diffusion, to analyze the speed of convergence to stationarity after an aggregate shock—which the spectral gap of the fractional KF operator governs and which subdiffusion slows—to compute the welfare cost of idiosyncratic income risk, and to evaluate the distributional effects of a lump-sum transfer financed by a wealth tax.

4 Implications

This research has implications along several dimensions. On the theoretical side, it establishes a rigorous link between the microeconomic behavior of individual earners—subdiffusive CTRW dynamics and non-Markovian waiting times—and macroeconomic distributional outcomes such as the stationary wealth distribution, the Gini coefficient, and Pareto tail exponents. The variational structure of Qing and Camilli (2018) provides a clean mathematical framework for existence, uniqueness, and comparative statics.

On the empirical side, the model provides a structural interpretation of the high kurtosis and asymmetric persistence documented in the SSA data (Guvenen et al. 2015). The subdiffusion exponent β is a structural parameter directly linked to observable moments of the earnings distribution, making it empirically identified and interpretable. The adaptation of the Achdou et al. (2022) algorithm to fractional operators, implemented in MFAX (Wibault et al. 2026), also provides a reusable computational toolkit for economists wishing to incorporate non-Markovian dynamics into continuous-time heterogeneous-agent models more broadly.

On the policy side, if subdiffusive dynamics generate more persistent poverty traps and slower adjustment to redistributive policy than standard models predict, then the implications for the optimal timing and design of transfers, insurance programs, and wealth taxes may be substantively different from those

derived under standard Markovian assumptions. Hopefully, in using this framework in combination with rich administrative data, this research will offer more accurate guidance on these questions.

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